Lab: test a claim — solution

Session 2 — Data management

Data Analysis in R using Agentic AI Assistants · Kevin Koehler · 30 October 2026

Since the end of the Cold War, internal armed conflict has been concentrated in poorer countries.

Short answer: yes. Between 1991 and 2025, a third (33%) of all country-years in the poorest quarter of countries saw an internal armed conflict, against 5% in the richest quarter. Almost half (48%) of all conflict country-years fall in the poorest quarter. The pattern holds in each decade, whether we use income in the same or the previous year, and it is even stronger for wars (1,000 or more battle deaths).

This document goes through the five steps of the lab. It shows one good way of doing each step, the numbers you should have found, and the places where things usually go wrong. Other reasonable choices exist; what matters is that you can say what you chose and why. All numbers come from two scripts, printed in full at the end:

- `scripts/lab2_merge.R`: builds `output/conflict_income.csv` (Steps 2–3);
- `scripts/lab2_test.R`: runs the test and draws the chart (Step 5).

They follow the course `AGENTS.md`: simple steps, a row count after every step that can change the number of rows.

Step 1: Find the data

Data used here: **UCDP/PRIO Armed Conflict Dataset, version 26.1** and the World Bank's **World Development Indicators**, indicator `NY.GDP.PCAP.PP.KD` (database last updated 8 October 2026).

	UCDP	**World Bank**
Dataset / indicator, version	UCDP/PRIO Armed Conflict Dataset, v. 26.1	GDP per capita, PPP (constant 2021 international $), `NY.GDP.PCAP.PP.KD`
File	`UcdpPrioConflict_v26_1.csv` (2,816 rows)	`API_NY.GDP.PCAP.PP.KD_DS2_en_csv_v2_1629.csv`, plus `Metadata_Country_...csv`
One row = ?	One **conflict in one year**. A conflict is a government against one incompatibility; several rebel groups can share a row (Colombia: "ELN, FARC").	One **country** (or region, or income group) with **all** years.
Country identified by	`location` (a name; old names in brackets, e.g. "Russia (Soviet Union)") and `gwno_loc`: **Gleditsch & Ward** numbers, stored as text	`Country Name` and `Country Code`: **ISO-3 letters** (`TUN`); aggregates have made-up codes (`WLD`, `ARB`)
Years	1946–2025	Columns 1960–2025; values only from 1990
Variables used	`type_of_conflict` (3 = intrastate, 4 = internationalized intrastate); `gwno_loc` (where); `intensity_level` (1 = 25–999 battle deaths in the year, 2 = 1,000+)	GDP per person, adjusted for price levels across countries (PPP) and over time (constant 2021 prices), in international dollars
Wide or long?	Long (one row per conflict-year)	**Wide** (one column per year)
Missing	A country-year **without conflict has no row**. Empty text (`""`) where a field does not apply, e.g. `side_a_2nd`	Empty cell. Four lines of notes above the header. Some countries have no values at all (e.g. Yemen, South Sudan, Venezuela, Eritrea)

The two questions in the small print

What counts as "armed conflict"? UCDP: a contested incompatibility over government and/or territory, in which armed force between two parties, at least one of them the government of a state, causes at least 25 battle-related deaths in a calendar year. **Internal** conflicts are types 3 (intrastate) and 4 (internationalized intrastate: an internal conflict in which other states send troops to one side). Type 1 (extrasystemic, colonial wars; none after 1974) and type 2 (interstate) are not internal.

Which GDP per capita? We compare across countries *and* across years, so we need **PPP** (removes differences in price levels between countries) and **constant prices** (removes inflation). "Current US$" mixes real income with exchange rates and inflation: a country can look poorer simply because its currency fell.

Step 2: Prompt the merge

The Step 1 table shows that three things must happen before a merge is possible:

- **Codes differ.** UCDP uses G&W numbers, the World Bank ISO-3 letters. One must be converted into the other (`countrycode`). Matching by *name* would fail for "Russia (Soviet Union)" vs "Russian Federation", "DR Congo (Zaire)" vs "Congo, Dem. Rep.", "Yemen (North Yemen)" vs "Yemen, Rep.", and others.
- **UCDP is not one row per country-year.** It is one row per *conflict*-year, so it must be

aggregated (counted) to country-years first.

- **The World Bank file is wide.** It must be reshaped to long (one row per country-year).

A further decision belongs in the prompt: which country-years make up the table? If we start from UCDP, we only have country-years *with* conflict, and nothing to compare them with. We start from the World Bank countries and add conflict information to them.

Model prompt

Write an R script `scripts/lab2_merge.R` that combines two datasets into one table with one row per country and year, 1990–2025, and saves it as `output/conflict_income.csv`.

Conflict: `data/UcdpPrioConflict_v26_1.csv` (UCDP/PRIO Armed Conflict Dataset). One row is one conflict in one year. Keep only internal conflicts (`type_of_conflict` 3 or 4) from 1990 on. The country where the conflict takes place is `gwno_loc`, a Gleditsch & Ward code. Convert it to ISO-3 letter codes with the `countrycode` package and show me any codes it cannot convert. Then count conflicts per country and year so that there is one row per country-year, and keep the highest `intensity_level`.

Income: `data/API_NY.GDP.PCAP.PP.KD_DS2_en_csv_v2_1629.csv` (World Bank, GDP per capita, PPP, constant 2021 international $). The first four lines are notes. It is wide, one column per year: make it long, with `year` as a number. The country is `Country Code` (ISO-3 letters). Drop regions and income groups: in `data/Metadata_Country_...csv` they have no `Region`.

Merge: keep all World Bank country-years and add the conflict information. A country-year with no UCDP row had no internal conflict: code it 0, not `NA`. Before merging, check that the key is unique in both tables, and list the conflict country-years that find no match.

The prompt names the files, the identifiers and code systems, the years, the unit of the final table, and the three preparation steps. It also asks for the checks we will need in Step 3.

What usually goes wrong here

A prompt such as "merge the UCDP and World Bank data by country and year" leaves every decision to the assistant. Typical results: a join on country *names*; all conflict types, including interstate wars; `gwno_a` (the government side) instead of `gwno_loc`; no aggregation, so country-years with several conflicts appear several times; an `inner_join()`, which keeps only conflict country-years; or `NA` instead of 0 for peace.

Step 3: Inspect the data

1. Did the merge work?

`nrow(d)` = **6,228** rows; **173** countries; years **1990–2025** (36 years). $173 \times 36 = 6{,}228$: every country has every year, exactly as it should. There are **1,094** country-years with an internal conflict, and all of them found their country in the World Bank table.

2. What did not match?

- **World Bank entries without a G&W code** (44 of 217): territories (Aruba, Puerto Rico, Hong Kong, Greenland, West Bank and Gaza) and very small states (Andorra, Monaco, Seychelles, Tonga, Samoa…). UCDP's list of states (G&W) does not include them, so they cannot have a conflict in the data. We drop them; keeping them would add country-years that are "peaceful" by construction.
- **UCDP codes that `countrycode` cannot convert:** `345` ("Serbia (Yugoslavia)") and `678` ("Yemen (North Yemen)"). G&W keep the old state's code for the successor. Unless you add them by hand, Yugoslavia/Serbia (Slovenia and Croatia 1991–92, Kosovo 1998–99) and **Yemen (19 conflict years)** silently disappear. In the other direction, `countrycode` does not know the G&W codes for Yemen (`YEM` → 678) or Kosovo (`XKX` → 347).
- **Country names**, if you merged on them: see Step 2.

3. Is it tidy?

Yes: `d %>% count(iso3c, year) %>% filter(n > 1)` returns no rows; every cell holds one value. Before aggregation, UCDP had **1,553** internal conflict-year rows for **1,094** country-years; 300 country-years had more than one conflict (India had six in 1994, 1996 and 1997). Joining those rows directly would have added 459 duplicate rows, each carrying the same income again.

4. Is everything there?

A country-year without conflict (Italy 2015) is in the table with `conflict = 0`. In the merge itself 5,134 country-years found no UCDP row and came out as `NA`; the script turns them into 0. That is right for our claim: UCDP lists *every* conflict that reached 25 battle deaths, so no row means "no conflict", not "unknown". With `NA`, those country-years would drop out of every share, and we would compare conflict country-years only with each other.

Income is a different matter: **287** country-years have no GDP. Some countries have no PPP series at all (Cuba, Eritrea, North Korea, South Sudan, Venezuela, Yemen); others start late (Kosovo, Djibouti, Afghanistan). Several of them are conflict countries. We come back to this in Step 5.

5. Spot-check by hand

Country	Year	GDP p.c.	Conflict	In the UCDP file
Tunisia	1995	7,406	0	no row
Tunisia	2015	12,803	1	Government vs AQIM, JAK-T; intensity 1
Tunisia	2025	13,018	0	no row
Egypt	1995	8,424	1	Government vs al-Gama'a al-Islamiyya
Egypt	2015	13,426	1	Government vs IS
Colombia	1995	11,279	1	Government vs ELN, FARC
Colombia	2015	16,821	1	Government vs ELN, FARC
Colombia	2025	18,779	1	Government vs ELN, EMBF, FARC-EMC

The values agree with the original files.

Step 4: Read the code

The questions refer to `scripts/lab2_merge.R` (Appendix A).

1. **Converting identifiers.** `countrycode(internal$gwno, origin = "gwn", destination = "iso3c", custom_match = c("345" = "SRB", "678" = "YEM"))` turns G&W numbers into ISO-3 letters. Codes it cannot convert become `NA`; `countrycode` prints a *warning* listing them, which is easy to overlook. The next line, `sum(is.na(internal$iso3c))`, makes the check visible (it prints 0). The same function, with `destination = "gwn"`, gives each World Bank country its G&W code.
2. **Aggregating and reshaping.** `group_by(iso3c, year) %>% summarise(n_conflicts = n(), ...)`: before, one row = one conflict-year (1,553 rows); after, one row = one country-year (1,094 rows). `pivot_longer()`: before, one row = one country with 66 year columns; after, one row = one country-year (14,322 rows for all 217 countries and 66 years, before filtering). Afterwards `year` is text and is turned into a number with `as.numeric()`.
3. **The merge.** `left_join(conflict_cy, by = c("iso3c", "year"))`: a *left* join keeps every row of the first table (the World Bank country-years) and adds the conflict columns where it finds a match. Rows without a match get `NA`, which we recode to 0. The `anti_join()` just before shows which conflict country-years would *not* find a match (none).
4. **The line you understand least.** Often `wb[, names(wb) != ""]` (keep all columns whose name is not empty: the World Bank file ends every line with a comma, which creates a nameless last column) or `names(wb_meta)[1] <- "Country.Code"` (the metadata file starts

with an invisible byte-order mark, which R reads into the first column's name).

Step 5: Design the test

Decisions

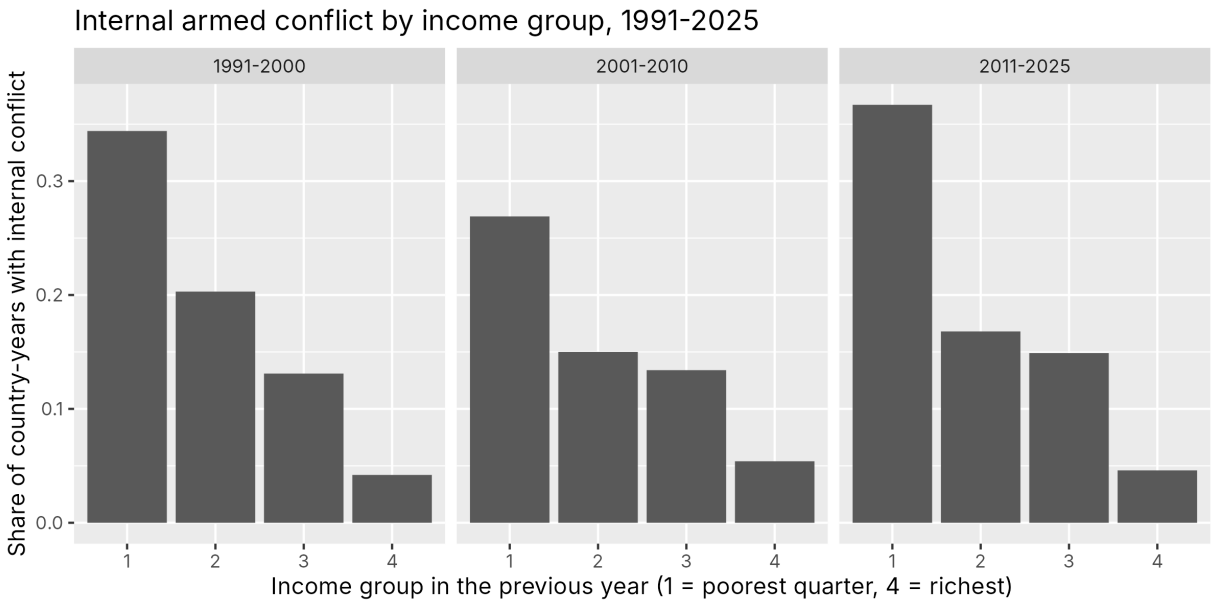
1. **"Concentrated":** conflict is much more common among poorer countries. We compare the share of country-years with conflict across income groups, and report how all conflict country-years are spread over those groups. A difference in averages (Test 2) is a second view.
2. **"Poorer":** four equal-sized income groups (quartiles), formed *within each year*. Everybody got richer between 1990 and 2025; pooled groups would put most early years into the poor groups and most late years into the rich ones. Within-year groups measure relative poverty: poorer than other countries at the time.
3. **Years:** 1991–2025. The Cold War ended 1989–91; UCDP's own convention is often 1989, but the World Bank PPP series starts in 1990 and we need the previous year's income.
4. **Reverse causation:** war destroys income, so a country that is poor *during* a war is weak evidence. We use income in the *previous* year. This only partly solves the problem (many conflicts last for years, so last year was a war year too); a stricter test would look at the *start* of conflicts (onset) and income before it.
5. **What would make the claim false:** conflict shares roughly equal across the four groups, or a share in the poorest group not clearly above the others (a common rule of thumb: less than twice the richest group's), or a pattern that appears only in one period.

Missing income. 276 of 6,055 country-years (1991–2025) have no income for the previous year; the script drops them and prints how many. They are missing slightly more often in conflict years (5.2%) than in peace years (4.4%). The conflict years lost are mostly Yemen (19), South Sudan (12) and Afghanistan (10), plus a few from Djibouti, Eritrea, Serbia and Syria. These are poor countries at war, so dropping them makes the test *harder* for the claim, not easier.

Results

Income group (previous year)	Country-years	With conflict	Share with conflict	Share of all conflict years
1 (poorest quarter)	1,460	485	33.2%	48.3%
2	1,451	251	17.3%	25.0%
3	1,440	201	14.0%	20.0%
4 (richest quarter)	1,428	67	4.7%	6.7%

- **Test 2:** the median income in the year before a conflict year is 4,870 international dollars, against 13,689 in the year before a peace year.
- **Test 3, by period** (figure below): the poorest quarter has 34%, 27% and 37% conflict country-years in 1991–2000, 2001–2010 and 2011–2025; the richest quarter 4–5% in every period.
- **Test 4, wars only** (1,000+ battle deaths): 9.0% of country-years in the poorest quarter, 0.5% in the richest.
- **Test 5, same-year income:** 33.4%, 17.6%, 13.4%, 4.8%. Using the previous year's income hardly changes anything.
- **Rich countries with conflict:** almost all 67 country-years are Israel (30; conflicts located in Israel and the Palestinian territories), the United States (18; the conflict with al-Qaida, which UCDP locates in the US) and Russia (16; mainly Chechnya and the North Caucasus).



At the end of class

Is the claim true? How sure are you?

As a description, yes, and quite clearly. Country-years in the poorest quarter are about seven times as likely to see internal conflict as those in the richest quarter, almost half of all conflict years fall in the poorest quarter, and this holds in every period and for every variant we tried. Two limits: this is a *description*, not a cause (poorer countries also differ in institutions, state capacity, geography, history); and conflict lasts, so last year's income is often itself shaped by war.

Which decisions changed the answer the most?

Here the direction of the result survives every choice, but the size moves. What counts as conflict matters most: for wars the gap between poorest and richest is 18 to 1 rather than 7 to 1. The decisions that could have *broken* the result are the data-management ones: `NA` instead of 0 for peace (no comparison group left), an `inner_join()` (the same), current instead of constant PPP dollars, groups pooled across years, or losing Yemen and Serbia in the code conversion.

Where did the assistant decide something you had not asked for?

Look for: which conflict types to keep; `gwno_a` versus `gwno_loc`; what to do with codes `countrycode` cannot convert (often: nothing, only a warning); `inner_join()` versus `left_join()`; `NA` versus 0; dropping missing income silently (`na.rm = TRUE`, `drop_na()`); keeping or dropping regional aggregates such as "World" or "Arab World", which otherwise sit in the table as if they were countries.

Appendix A: scripts/lab2_merge.R

```r
# Lab 2, Steps 2-3: one table with one row per country and year
# Conflict: UCDP/PRIO Armed Conflict Dataset v26.1 (one row per conflict-year)
# Income: World Bank, GDP per capita, PPP (constant 2021 international $), wide

library(dplyr)
library(tidyr)
library(countrycode)


# ---- 1. Conflict data ----

ucdp <- read.csv("data/UcdpPrioConflict_v26_1.csv")
nrow(ucdp)

# Keep internal conflicts: 3 = intrastate, 4 = internationalized intrastate
internal <- ucdp %>% filter(type_of_conflict == 3 | type_of_conflict == 4)
nrow(internal)

# Keep the years after the Cold War
internal <- internal %>% filter(year >= 1990)
nrow(internal)

# gwno_loc is the Gleditsch & Ward code of the country where the conflict takes place.
# It is stored as text; for internal conflicts it holds a single code.
internal$gwno <- as.numeric(internal$gwno_loc)
sum(is.na(internal$gwno))

# Translate G&W codes into ISO-3 letter codes, which the World Bank uses.
# countrycode cannot translate 345 (Yugoslavia, used by UCDP for Serbia)
# and 678 (Yemen), so we give these two by hand.
internal$iso3c <- countrycode(internal$gwno, origin = "gwn", destination = "iso3c",
                              custom_match = c("345" = "SRB", "678" = "YEM"))

# Any codes still not translated?
sum(is.na(internal$iso3c))

# One row is one conflict in one year. Some countries have several conflicts
# in the same year, so we count them: one row per country-year.
conflict_cy <- internal %>%
  group_by(iso3c, year) %>%
  summarise(n_conflicts = n(),
            max_intensity = max(intensity_level),
            .groups = "drop")
nrow(conflict_cy)

# How many country-years had more than one conflict?
sum(conflict_cy$n_conflicts > 1)


# ---- 2. Income data ----

# The first four lines of the World Bank file are notes, so we skip them
wb <- read.csv("data/API_NY.GDP.PCAP.PP.KD_DS2_en_csv_v2_1629.csv", skip = 4, check.names =
    FALSE)
nrow(wb)

# The file ends with an empty column without a name; drop it
wb <- wb[, names(wb) != ""]

# The file also contains regions and income groups ("Arab World", "World").
# In the metadata file these rows have no Region, so we keep only rows with a Region.
wb_meta <- read.csv("data/Metadata_Country_API_NY.GDP.PCAP.PP.KD_DS2_en_csv_v2_1629.csv")
# The file starts with an invisible character that ends up in the name of the
```

```r
# first column ("X...Country.Code"), so we rename that column
names(wb_meta)
names(wb_meta)[1] <- "Country.Code"
countries <- wb_meta %>% filter(Region != "")
nrow(countries)

wb <- wb %>% filter(`Country Code` %in% countries$Country.Code)
nrow(wb)

# Keep the name, the code and the year columns
wb <- wb %>% select(-`Indicator Name`, -`Indicator Code`)

# Wide to long: one row per country and year
wb_long <- wb %>%
  pivot_longer(cols = -c(`Country Name`, `Country Code`),
               names_to = "year",
               values_to = "gdppc")
nrow(wb_long)

# Year is text after pivot_longer; make it a number
wb_long$year <- as.numeric(wb_long$year)

# Clear names
wb_long <- wb_long %>% rename(country = `Country Name`, iso3c = `Country Code`)

# Keep 1990 to 2025 (the World Bank series starts in 1990)
wb_long <- wb_long %>% filter(year >= 1990 & year <= 2025)
nrow(wb_long)

# UCDP only covers states in the Gleditsch & Ward list. Some World Bank "countries"
# are not on it: territories (e.g. Aruba, Puerto Rico, Hong Kong) and very small
# states. We give each country its G&W code and drop those without one.
# countrycode does not know the codes for Yemen (678) and Kosovo (347): we add them.
wb_long$gwno <- countrycode(wb_long$iso3c, origin = "iso3c", destination = "gwn",
                            custom_match = c("YEM" = 678, "XKX" = 347))
unique(wb_long$country[is.na(wb_long$gwno)])
wb_long <- wb_long %>% filter(!is.na(gwno))
nrow(wb_long)


# ---- 3. Merge ----

# Is the key unique in both tables? (Both should print 0.)
conflict_cy %>% count(iso3c, year) %>% filter(n > 1) %>% nrow()
wb_long %>% count(iso3c, year) %>% filter(n > 1) %>% nrow()

# Which conflict country-years will find no partner in the World Bank table?
conflict_cy %>% anti_join(wb_long, by = c("iso3c", "year"))

# Left join: keep every World Bank country-year, add the conflict information
d <- wb_long %>% left_join(conflict_cy, by = c("iso3c", "year"))
nrow(d)

# A country-year without a UCDP row had no internal conflict: that is 0, not NA
sum(is.na(d$n_conflicts))
d$n_conflicts[is.na(d$n_conflicts)] <- 0
d$max_intensity[is.na(d$max_intensity)] <- 0

# Conflict yes/no
d$conflict <- ifelse(d$n_conflicts > 0, 1, 0)

# Save
d <- d %>% select(country, iso3c, gwno, year, gdppc, conflict, n_conflicts, max_intensity)
write.csv(d, "output/conflict_income.csv", row.names = FALSE)
```

Appendix B: scripts/lab2_test.R

```
# Lab 2, Step 5: is internal armed conflict concentrated in poorer countries?
# Test: share of country-years with an internal conflict, by income group,
# using income in the PREVIOUS year (war lowers income, not only the reverse).

library(dplyr)
library(ggplot2)

d <- read.csv("output/conflict_income.csv")
nrow(d)

# Income in the previous year, within each country
d <- d %>%
  arrange(iso3c, year) %>%
  group_by(iso3c) %>%
  mutate(gdppc_lag = lag(gdppc)) %>%
  ungroup()

# 1990 has no previous year in our data, so the test starts in 1991
d <- d %>% filter(year >= 1991)
nrow(d)


# ---- Missing income ----

# How many country-years have no income for the previous year?
sum(is.na(d$gdppc_lag))

# Is income missing more often in conflict years? (share missing, by conflict)
d %>%
  group_by(conflict) %>%
  summarise(country_years = n(),
            missing_income = sum(is.na(gdppc_lag)),
            share_missing = round(mean(is.na(gdppc_lag)), 3))

# Which conflict countries lose the most years?
d %>%
  filter(conflict == 1 & is.na(gdppc_lag)) %>%
  count(country, sort = TRUE)

# Drop country-years without income, and say how many
d_test <- d %>% filter(!is.na(gdppc_lag))
nrow(d) - nrow(d_test)
nrow(d_test)


# ---- Income groups ----

# Four groups of equal size within each year, from 1 = poorest to 4 = richest.
# Within each year, because all countries got richer between 1991 and 2025.
d_test <- d_test %>%
  group_by(year) %>%
  mutate(income_group = ntile(gdppc_lag, 4)) %>%
  ungroup()

# Check: roughly equal numbers of country-years per group
table(d_test$income_group)


# ---- Test 1: share of country-years with conflict, by income group ----

by_group <- d_test %>%
  group_by(income_group) %>%
  summarise(country_years = n(),
```

```
            with_conflict = sum(conflict),
            share_conflict = round(mean(conflict), 3))
by_group

# How are all conflict country-years distributed across the four groups?
d_test %>%
  filter(conflict == 1) %>%
  count(income_group) %>%
  mutate(share = round(n / sum(n), 3))


# ---- Test 2: typical income with and without conflict ----

# Median rather than mean: a few very rich countries pull the mean up
d_test %>%
  group_by(conflict) %>%
  summarise(country_years = n(),
            median_gdppc_lag = round(median(gdppc_lag)))


# ---- Test 3: does the pattern hold in each period? ----

d_test$period <- ifelse(d_test$year <= 2000, "1991-2000",
                 ifelse(d_test$year <= 2010, "2001-2010", "2011-2025"))

by_period <- d_test %>%
  group_by(period, income_group) %>%
  summarise(country_years = n(),
            share_conflict = round(mean(conflict), 3),
            .groups = "drop")
by_period


# ---- Test 4: only wars (1,000 or more battle deaths in the year) ----

d_test$war <- ifelse(d_test$max_intensity == 2, 1, 0)

d_test %>%
  group_by(income_group) %>%
  summarise(country_years = n(),
            share_war = round(mean(war), 3))


# ---- Test 5: income in the same year instead of the previous year ----

d_same <- d %>% filter(!is.na(gdppc))
nrow(d_same)

d_same <- d_same %>%
  group_by(year) %>%
  mutate(income_group_same = ntile(gdppc, 4)) %>%
  ungroup()

d_same %>%
  group_by(income_group_same) %>%
  summarise(country_years = n(),
            share_conflict = round(mean(conflict), 3))


# ---- Which rich countries have conflicts? ----

d_test %>%
  filter(income_group == 4 & conflict == 1) %>%
  count(country, sort = TRUE)
```

```r
# ---- Chart ----

p <- ggplot(by_period, aes(x = factor(income_group), y = share_conflict)) +
  geom_col() +
  facet_wrap(~ period) +
  labs(title = "Internal armed conflict by income group, 1991-2025",
       x = "Income group in the previous year (1 = poorest quarter, 4 = richest)",
       y = "Share of country-years with internal conflict")
p

ggsave("output/conflict_by_income.png", p, width = 8, height = 4)
```